11 – AI tools (Tool Use)

Definition: With **tools**, an AI can do more than write text: for example, it calls a web search, a calculator, or a file analysis and uses the result in its answer.

> **Remember:** With tools, the AI doesn't just talk – it acts.

Key points

- **Examples:** web search, file analysis, code execution (calculating, charts), image generation, connecting to calendar or mail.
- **Principle:** The model decides that it needs a tool, calls it, and builds the result into its answer. You often see this as a note like "I'm searching the web …".
- **When tools help:** with current information (web search), with exact calculations and data evaluation (code execution), with your own documents (file analysis).
- **What changes:** With tools, the AI can also *do* things, not just write – for example, enter an appointment. That creates new risks (Block C).
- **Not available everywhere:** Which tools your service offers depends on the service and the plan (topic 03). Check the current details with your provider.
- **Check access rights:** Grant only what you need for the task. Read what you allow.
- **Checking remains necessary:** Even a tool can be used wrongly – for example, when the AI uses the wrong column of a table.

Example

You want to know where your money goes. The AI should calculate, not estimate:

```
Here are my expenses as a table. Calculate the total per category
and create a bar chart. Tell me how you calculated.
```

Without a calculation tool, an AI can get sums wrong. With code execution, it usually calculates more reliably – but you still spot-check.

Do / Don't

Do	Don't
Use tools for current info and calculations	Believe current information without web search
Grant only necessary access	Give full access to mail and files "for safety"
Confirm actions with consequences yourself	Let sending, deleting, or paying run automatically

Safety note

What the AI reads via tools (web pages, files, mails) can contain hidden instructions (topic 12). Don't upload files with sensitive data (topic 04). Always confirm actions yourself.

Mini checklist

- ☐ I know at least three tools.
- ☐ I know which tools my service offers.
- ☐ I grant only necessary access.
- ☐ I spot-check results.

Exercise (5 min.)

Upload this made-up table as a `.csv` file. If upload isn't possible, paste it into the chat. Use the prompt above.

```
Week,Category,Amount
1,Groceries,85
1,Leisure,30
2,Groceries,92
2,Leisure,15
3,Groceries,78
3,Leisure,40
```

Check the spot-check: the total for groceries must be 255, and the total for leisure 85.

Next

- Extra exercises: `02_extra_exercises/11_extra_exercises`
- Test: Test B (Good results)
- Related: 01 What is AI?, 08 Hallucinations, 12 Prompt injection, 16 Agents

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